§2.4

2.4.0. Decide which of the following statements are true and which are false. Prove the true ones and provide a counterexample for the false ones.

a) If $\{x_n\}$ is Cauchy and $\{y_n\}$ is bounded, then $\{x_n y_n\}$ is Cauchy.
b) If $\{x_n\}$ and $\{y_n\}$ are Cauchy and $y_n \neq 0$ for all $n \in \mathbf{N}$, then $\{x_n/y_n\}$ is Cauchy.
c) If $\{x_n\}$ and $\{y_n\}$ are Cauchy and $x_n + y_n > 0$ for all $n \in \mathbf{N}$, then $\{1/(x_n + y_n)$ cannot converge to zero.

d) If $\{x_n\}$ is a sequence of real numbers that satisfies $x_{2^k} - x_{2^{k-1}} \to 0$ as $k \to \infty$ and if $x_n = 0$ for all $n \neq 2^k$, $k \in \mathbf{N}$, then $\{x_n\}$ is Cauchy.

(a) False!

Let $x_n = 1^n$ is convergent, $y_n = (-1)^n$ is bounded.

Since x_n is convergent, x_n is Cauchy.

We have $x_n y_n = (-1)^n$, $x_n y_n$ does not conv.

$\Rightarrow x_n y_n$ is not Cauchy. //

(d) False!

Let $x_n = \begin{cases} \log k, & \text{if } n = 2^k,\ k \in \mathbb{N}. \\ 0, & \text{if } n \neq 2^k, \end{cases}$

then $x_{2^k} - x_{2^{k-1}} = \log k - \log(k-1)$

$= \log \frac{k}{k-1} \to 0$ as $k \to \infty$

and $x_n \to 0$ as $n \neq 2^k \to \infty$.

But $x_n \to \infty$ as $n = 2^k \to \infty$, x_n does not conv.

$\Rightarrow x_n$ is not Cauchy.

2.4.1. Prove that if $\{x_n\}$ is a sequence that satisfies

$$|x_n| \leq \frac{5n^2+3n+7}{n^4+4n^2-1}$$

for all $n \in \mathbf{N}$, then $\{x_n\}$ is Cauchy.

pf $|x_n| \leq \frac{5n^2+3n+7}{n^4+4n^2-1} \Rightarrow -\frac{5n^2+3n+7}{n^4+4n^2-1} \leq x_n \leq \frac{5n^2+3n+7}{n^4+4n^2-1}$

Since $-\frac{5n^2+3n+7}{n^4+4n^2-1} \to 0$ and $\frac{5n^2+3n+7}{n^4+4n^2-1} \to 0$ as $n \to \infty$,

by the Squeeze Thm, $x_n \to 0$ as $n \to \infty$

$\Rightarrow x_n$ conv.

$\Rightarrow x_n$ is Cauchy. //

2.4.4. Let $\{x_n\}$ be a sequence of real numbers. Suppose that for each $\varepsilon > 0$ there is an $N \in \mathbf{N}$ such that $m \geq n \geq N$ implies $\left|\sum_{k=n}^{m} x_k\right| < \varepsilon$. Prove that

$$\lim_{n\to\infty} \sum_{k=1}^{n} x_k$$

exists and is finite.

pf ① If $m=n$, $\left|\sum_{k=n}^{m} x_k\right| = 0 < \varepsilon$.

② If $m>n$, let $S_n = \sum_{k=1}^{n} x_k$, then $\sum_{k=n}^{m} x_n = S_m - S_{n-1}$

This implies $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t.

$$\left|\sum_{k=n}^{m} x_k\right| = |S_m - S_{n-1}| < \varepsilon \text{ as } n, m \geq N+1$$

$\Rightarrow S_n$ is Cauchy

$\Rightarrow S_n$ conv.

$\Rightarrow \lim_{n\to\infty} \sum_{k=1}^{n} x_k$ exists and is finite. //

§3.1

3.1.0. Let $a \in \mathbf{R}$ and let f and g be real functions defined at all points x in some open interval containing a except possibly at $x = a$. Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples for the false ones.

a) For each $n \in \mathbf{N}$, the function $(x-a)^n \sin(f(x)(x-a)^{-n})$ has a limit as $x \to a$.

b) Suppose that $\{x_n\}$ is a sequence converging to a with $x_n \neq a$. If $f(x_n) \to L$ as $n \to \infty$, then $f(x) \to L$ as $x \to a$.

c) If f and g are finite valued on the open interval $(a-1, a+1)$ and $f(x) \to 0$ as $x \to a$, then $f(x)g(x) \to 0$ as $x \to a$.

d) If $\lim_{x\to a} f(x)$ does not exist and $f(x) \leq g(x)$ for all x in some open interval I containing a, then $\lim_{x\to a} g(x)$ doesn't exist either.

(a) True

pf

$$-1 \leq \sin\left(f(x)(x-a)^{-n}\right) \leq 1$$

$$\Rightarrow -(x-a)^n \leq (x-a)^n \sin\left(f(x)(x-a)^{-n}\right) \leq (x-a)^n$$

Since $-(x-a)^n \to 0$ and $(x-a)^n \to 0$ as $x \to a$,

by the Squeeze Thm, $(x-a)^n \sin\left(f(x)(x-a)^{-n}\right) \to 0$ as $x \to a$.

(d) False!

pf Let $f(x) = \sin\left(\frac{1}{x}\right)$ and $g(x) = 1$.

It is clear that $\sin\left(\frac{1}{x}\right) = f(x) \leq g(x) = 1$

and $f(x)$ DNE as $x \to 0$.

But $\lim_{x\to 0} g(x) = 1$ ~~$\neq$~~ //

3.1.1. Using Definition 3.1, prove that each of the following limits exist.

a) $\lim_{x\to 1} x^2 + 5x + 6 = 12$

d) $\lim_{x\to 0} x\cos(\pi x) = 0$

(a) $\lim_{x\to 1} x^2+5x+6 = 12$

pf Given $\varepsilon>0$, take $\delta = \min\{1, \frac{\varepsilon}{8}\}$

Since $|x-1|<\delta \le 1 \Rightarrow 0<x<2$

$\Rightarrow 6 < x+6 < 8$

$\Rightarrow |x+6| < 8$

If $0<|x-1|<\delta \le \frac{\varepsilon}{8}$, then

$|x^2+5x+6-12| = |x^2+5x-6|$

$= |x+6||x-1|$

$< 8 \cdot \frac{\varepsilon}{8} = \varepsilon$

Hence $\lim_{x\to 1} x^2+5x+6 = 12$. //

$\forall \varepsilon>0, \exists \delta>0$ s.t.

$|x^2+5x+6-12| \le \frac{\varepsilon}{8}$

$= |x^2+5x-6| \le \varepsilon$

$= |x+6||x-1| < 8\cdot\delta$

Take $\delta \le 1$, since

$|x-1|<1 \Rightarrow 0<x<2$

$\Rightarrow 6<x+6<8 \Rightarrow |x+6|<8$

take $\delta = \min\{1, \frac{\varepsilon}{8}\}$

(d) $\lim_{x\to 0} x\cos(\pi x) = 0$.

pf Given $\varepsilon>0$, take $\delta=\varepsilon$.

Since $|\cos(\pi x)| \le 1$, $\forall x\in\mathbb{R}$,

if $0<|x|<\delta$, then $|x\cos(\pi x)| = |x||\cos(\pi x)|$

$< \delta \cdot 1 = \varepsilon$

Hence $\lim_{x\to 0} x\cos(\pi x) = 0$. //

3.1.3. Evaluate the following limits using results from this section. (You may assume that $\sin x$, $1-\cos x$, $\tan x$, and $\sqrt[4]{x}$ converge to 0 as $x \to 0$.)

a) $\lim_{x\to -1} \dfrac{x^2+6x+5}{x^4-x^2}$

Sol. $\lim_{x\to -1} \dfrac{x^2+6x+5}{x^4-x^2} = \lim_{x\to -1} \dfrac{(x+1)(x+5)}{x^2(x+1)(x-1)} = \dfrac{4}{1\cdot -2} = -2$ //

3.1.6. Suppose that f is a real function.

a) Prove that if

$$L = \lim_{x\to a} f(x)$$

exists, then $|f(x)| \to |L|$ as $x \to a$.

b) Show that there is a function such that, as $x \to a$, $|f(x)| \to |L|$ but the limit of $f(x)$ does not exist.

(b) If $L \neq 0$,

let $f(x) = \begin{cases} L & \text{if } x \geq a \\ -L & \text{if } x < a \end{cases}$, then $|f(x)| = |L|$, $\forall x \in \mathbb{R}$.

It is clear that $|f(x)| \to |L|$ as $x \to a$, but $\lim_{x\to a} f(x)$ DNE. //

§3.2

3.2.1. For each of the following, use definitions (rather than limit theorems) to prove that the limit exists. Identify the limit in each case.

b) $\lim_{x\to\infty} \dfrac{\cos x}{x} = 0$

$\forall \varepsilon > 0, \exists M \in \mathbb{R}$ s.t.
$\left|\frac{\cos x}{x} - 0\right| < \varepsilon$ as $x > M$
$\left|\frac{\cos x}{x}\right| \le \frac{1}{|x|} < \frac{1}{M} = \varepsilon$
take $M = \frac{1}{\varepsilon}$

pf Given $\varepsilon > 0$, take $M = \frac{1}{\varepsilon}$.
We know $|\cos x| \le 1$, $\forall x \in \mathbb{R}$.
If $x > M$, then
$\left|\frac{\cos x}{x}\right| \le \frac{1}{|x|} < \frac{1}{M} = \varepsilon$
Hence $\lim_{x\to\infty} \frac{\cos x}{x} = 0$. //

3.2.2. Assuming that $e^x \to e^a$, $\sin x \to \sin a$, and $\cos x \to \cos a$ as $x \to a$ for any $a \in \mathbf{R}$, evaluate the following limits when they exist.

b) $\lim_{x\to\infty} \dfrac{3x^4 + x^2 - 5}{2x^4 + 2x^3 - 1}$

$$\lim_{x\to\infty} \frac{3x^4+x^2-5}{2x^4+2x^3-1} = \lim_{x\to\infty} \frac{3 + \frac{1}{x^2} - \frac{5}{x^4}}{2 + \frac{1}{x} - \frac{1}{x^4}} = \frac{3}{2}$$

§ 3.3

3.3.0. Decide which of the following statements are true and which are false. Prove the true ones and provide counterexamples for the false ones.

a) If f is continuous on $[a,b]$ and $J := f([a,b])$, then J is a closed, bounded interval.

b) If f and g are continuous on $[a,b]$, if $f(a) < g(a)$ and $f(b) > g(b)$, then there is a $c \in [a,b]$ such that $f(c) = g(c)$.

c) Suppose that f and g are defined and finite valued on an open interval I which contains a, that f is continuous at a, and that $f(a) \neq 0$. Then g is continuous at a if and only if fg is continuous at a.

d) Suppose that f and g are defined and finite valued on $\mathbf{R}$. If f and $g \circ f$ are continuous on $\mathbf{R}$, then g is continuous on $\mathbf{R}$.

(c) True

($\Rightarrow$) By Thm 3.22, fg is cont. at a.

($\Leftarrow$) If fg is cont. at a, let $g(x) = \dfrac{f(x)g(x)}{f(x)}$

Since $f(a) \neq 0$, $g(x) = \dfrac{f(x)g(x)}{f(x)}$ is cont. at a. //

(d) False

Let $f(x) = 0$ and $g(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$

Then $g \circ f(x) = g(f(x)) = g(0) = 0$, $\forall x \in \mathbb{R}$.

It is clear that f and $g \circ f$ are cont. on $\mathbb{R}$,

but g is not cont. at 0. //

3.3.1. Use limit theorems to show that the following functions are continuous on [0,1].

a) $f(x) = \dfrac{\sqrt{e^x}\cos(x^2+2x+6)}{\sin(x+10)}$

b) $f(x) = \begin{cases} \dfrac{3x^2-5x-2}{x-2} & x \neq 1 \\ 4 & x = 1 \end{cases}$

(b) If $x \in [0,1)$, since $3x^2-5x-2$ and $x-2$ are cont. on $\mathbb{R}$, $\dfrac{3x^2-5x-2}{x-2}$ is cont. on $[0,1)$.

Thus $f(x) = \dfrac{3x^2-5x-2}{x-2}$ is cont. on $[0,1)$.

If $x=1$, $\lim_{x\to 1} \dfrac{3x^2-5x-2}{x-2} = 4 = f(1) \Rightarrow f$ is cont. at $x=1$.

Hence f is cont. on $[0,1]$. //